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RESEARCH

Evaluation of Container Closure System Integrity for Frozen Storage Drug Products

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ABSTRACT: Sometimes, drug product for parenteral administration is stored in a frozen state (e.g., –20 °C or – 80 °C), particularly during early stages of development of some biotech molecules in order to provide sufficient stability. Shipment of frozen product could potentially be performed in the frozen state, yet possibly at different temperatures, for example, using dry ice (– 80 °C). Container closure systems of drug products usually consist of a glass vial, rubber stopper, and an aluminum crimped cap. In the frozen state, the glass transition temperature (T_g) of commonly used rubber stoppers is between –55 and – 65 °C. Below their T_g, rubber stoppers are known to lose their elastic properties and become brittle, and thus potentially fail to maintain container closure integrity in the frozen state. Leaks during frozen temperature storage and transportation are likely to be transient, yet, can possibly risk container closure integrity and lead to microbial contamination. After thawing, the rubber stopper is supposed to re-seal the container closure system. Given the transient nature of the possible impact on container closure integrity in the frozen state, typical container closure integrity testing methods (used at room temperature conditions) are unable to evaluate and thus confirm container closure integrity in the frozen state. Here we present the development of a novel method (thermal physical container closure integrity) for direct assessment of container closure integrity by a physical method (physical container closure integrity) at frozen conditions, using a modified He leakage test. In this study, different container closure systems were evaluated with regard to physical container closure integrity in the frozen state to assess the suitability of vial/stopper combinations and were compared to a gas headspace method. In summary, the thermal physical container closure integrity He leakage method was more sensitive in detecting physical container closure integrity impact than gas headspace and aided identification of an unsuitable container closure system.

KEYWORDS: Container closure integrity, Frozen product, Container closure system, Vial, Headspace analysis, He leakage test.

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Introduction

In certain cases, drug product for parenteral administration is stored in the frozen state (e.g., –20 °C, –80 °C) to overcome limitations of storage at 2–8 °C, which is usually the intended storage temperature for most biologic drug products. A few marketed products are also stored and/or shipped under frozen conditions (Table I). Storage of sterile pharmaceutical product vials at –80 °C has been used to ensure sufficient stability, particularly for live viral vaccines (1). Shipment of frozen product may be performed at different temperatures, for example, using dry ice (–80 °C). If these products are stored in glass vials with rubber stoppers, the elastomer (stopper) requires suitable flexibility to seal the vial at the low temperatures (at intended and lowest storage temperature, i.e., well below –15 °C) to ensure container closure integrity (CCI) and therefore the sterility and safety of the drug product. Hence, close consideration should be given to the assessment of CCI of a container closure system (CCS) during frozen storage and transportation.

To ensure quality and safety, the suitability of any frozen drug product strategy should be carefully assessed. Considerations that should be taken into account include

- Stability of product upon freezing and thawing.
- Process variability and consistency of product after freezing and thawing.
- Thawing process availability and reproducibility of thawing product (e.g., on clinical site).
- Capability to store and handle frozen product in a clinical hospital or study center.
- The ability to inspect product prior to use, for example, for particulates, turbidity, or any other liability from the freezing/thawing process given that handling (thawing) is performed prior administration.
- Solution behavior, for example, "liquid" frozen products may undergo a phase transition and may or may not expand (especially if containing mannitol) (2).
- CCI at frozen storage conditions. Given that the glass transition temperature (T_g) of standard stoppers is in the range of –50 to –65 °C (3), and even though the elasticity is regained by bringing the stopper back to room temperature (RT), the low temperature may lead to shrinkage and loss of elastomeric properties of the rubber with subsequent ingress of (unsterile) contaminant and subsequent loss of sterility (4, 5). Hence, the impact of low temperatures on the CCS may have an impact on CCI.

The *European Pharmacopoeia* clearly states that in all cases the container and closure are required to maintain the sterility of the product throughout its entire shelf life (6, 7). Sterility testing is not considered sufficient to demonstrate the microbial integrity of a CCS (8). Because sterility testing is not considered sufficient to ensure CCI of the drug product (9), sterility testing cannot replace a CCI assessment particularly during CCS qualification. It is imperative that the CCS is evaluated and proven, relative to CCI, to ensure maintenance of sterility. Any breach in integrity could lead to a serious risk to the product and to patients or healthy volunteers. Of note, this requirement exists not only for commercial drug products but needs consideration whenever human treatment is performed, that is, it also need to be considered prior clinical testing phases 1, 2, and 3.

In particular, the breach of CCI at low temperatures during frozen storage and shipment could lead to:

- Microbiological ingress, which could compromise the sterility of the product (4).
- Carbon dioxide (CO_2) ingress. If dry ice is used for storage and shipment, CO_2 ingress could lead to pH changes, possibly affecting the quality and efficacy of the product (1, 10–12).
- Oxygen (O_2) ingress, which could cause increased oxidation of the active pharmaceutical ingredient (API) and possibly could affect quality and efficacy (13).
- Loss of vacuum or overpressure in the headspace, which could lead to an increased level of impurities or even to new types of impurities as well as potency changes, and thus also possibly affect quality and efficacy (14).

Several considerations should be taken into account when evaluating CCI in the frozen state:

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TABLE I
Marketed Drug Products Stored and Shipped at Low Temperatures (29, 30)

Product Name	Active Ingredient	Indication	Therapeutic Modality	Storage Temperature (°C)	Presentation	Dosage Form	Excipients
Varivax	Weakened form of chickenpox virus	Varicella	Vaccine	–15 to –50	Glass vial	Powder	Sucrose; hydrolysed gelatin; urea; sodium chloride; monosodium L-glutamate; anhydrous disodium phosphate; potassium dihydrogen phosphate; potassium chloride
Zostavax	Attenuated (Oka-strain) varicella virus	Zoster	Vaccine	–15 to –50[a]	Glass vial	Powder	Sucrose; hydrolysed porcine gelatin; sodium chloride; monosodium L-glutamate; anhydrous disodium phosphate; potassium dihydrogen phosphate, potassium chloride; sodium hydroxide (pH adjustment)
ProQuad	Weakened strains of measles, mumps, rubella (German measles), and varicella viruses	MMRV	Vaccine	–15 to –50	Glass vial	Powder	Sucrose; hydrolysed gelatin; sodium chloride; sorbitol; monosodium glutamate; sodium phosphate; sodium bicarbonate; potassium phosphate; potassium chloride; Medium 199 with Hanks' Salts; minimum essential medium, Eagle (MEM); neomycin; phenol red; hydrochloric acid (to adjust pH); sodium hydroxide (to adjust pH); urea
M-M-R® II	Attenuated measles, mumps and rubella virus	MMRV	Vaccine	+8 to –50[b]	Glass vial	Powder	Sorbitol; sodium phosphate; sucrose; sodium chloride; hydrolyzed gelatine; recombinant human albumin; fetal bovine serum; neomycin; Medium 199 containing SPGA (sucrose, phosphate, glutamate, and recombinant human albumin)
Glybera	Alipogene tiparvovec	LPLD	Viral vector	–15 to –25	Glass vial	Solution for injection	Calcium chloride dihydrate; disodium phosphate anhydrous; magnesium chloride hexahydrate; potassium chloride; potassium dihydrogen phosphate; sodium chloride; sucrose

MMRV: Measles, mumps, rubella and varicella.
LPLD: Lipoprotein lipase deficiency.
[a] Shipped on dry ice.
[b] Use of dry ice may subject Measles, Mumps, and Rubella Virus Vaccine Live (M-M-R II®) to temperatures colder than –50 °C (1).

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- The CCI testing should use the "frozen" product and not the product after thawing, given that the thawing procedure may "re-seal" any leak that existed in the frozen state.

- The assurance of CCI needs to consider variability of processes, packaging materials, and the probabilistic nature of leaks, as well as microbiological ingress.

- The CCI assessment in frozen state should include: CCI of the CCS itself; the impact of processing, for example, the capping process and pressure on CCI (10), and impact of shipment and movement on CCI in frozen state (11).

A variety of methods can be used to evaluate the CCI of a CCS under ambient or refrigerated (non-frozen) conditions (15-18). Most methods of container closure integrity testing such as dye ingress, vacuum decay, or microbial ingress are, for various reasons, simply not feasible to be used for frozen product under frozen conditions.

Helium (He) leak detection is a very sensitive technique to evaluate physical container closure integrity (pCCI) (19). In addition, He leak detection provides quantitative results that are deterministic, reproducible, and more accurate than qualitative pass/fail results, although it is a destructive and, in general, lower-throughput test.

To date, only a few methodologies have been suggested to evaluate CCI of frozen drug products, including gas headspace analysis and a He leakage method (20–23).

The gas headspace method for CCI testing of frozen product uses laser-based frequency-modulated spectroscopy (FMS) to monitor headspace gas concentration changes over a period of time. The laser-based platforms enable rapid non-destructive characterization of the headspace gas conditions inside a sealed container. In a series of studies, crimped vials filled with He were placed in dry ice to allow ingress of carbon dioxide. The headspace gas concentration percentage (% atm) of CO_2 and O_2, and the headspace concentration were subsequently analyzed. Any increase in CO_2 in the vials would mean a breach in CCI (4). A major advantage of the method used by the authors is its non-destructive nature. Major limitations of the gas headspace method for CCI testing of frozen products include (i) limited applicability to containers without headspace volume; (ii) in the case of containers with atmospheric headspace, samples may need to be stored in special storage conditions (such as a nitrogen-rich environment) to allow for oxygen exchange (15); (iii) it is indirect, as the measurement takes place at RT after vials have been thawed (no direct CCI assessment at intended storage); (iv) longer turnaround times are needed for sensitive detection if the leaks are small; (v) lack of cross-correlation of pCCI to microbiological ingress testing (mCCI).

The He leakage method described in Cummings and Jacobs (22) consists of displacing the air in crimped vials with He by puncturing the stopper with a needle. Subsequently, the puncture sites are closed with epoxy resin. A chiller is connected to a He leak tester and vials are individually analyzed. The instrument places the sample under high vacuum and measures the He as it escapes from the sample vial. The He partial pressure present in the leak detector is measured by the mass spectrometer and displays this "leak" from the sample vial as a leak rate, which is measured quantitatively. After the leak rate has been determined, the vials are allowed to warm to RT. At this point, the He concentration inside the vial is measured with a calibrated headspace analyzer probe to determine the amount of He remaining in the vial. The He concentration and the measured He leak rate are then used to calculate the actual He leak rate of the vials, in units of standard atmosphere cubic centimeters per second (std.atm.cc/s). One millibar (mbar) $\times$ L/s is equal to 0.987 std.atm.cc/s. With this method, the authors were able to detect critical leak rates of 3.9×10^{-6} std.atm.cc/s at −80 °C and determined that a leak rate that is $<3 \times 10^{-6}$ std.atm.cc/s shows that the vial is *integer*. Because He was filled at RT and detected at low temperature, it was assumed that the pressure of He in the vial would be reduced at low temperature. Hence, the leak rates needed to be corrected to account for pressure reduction by using the ideal gas law. This was performed under the assumption that at low temperature the critical leak rate would be reduced in direct proportion to the reduction in internal vial pressure.

A diagram of the He leakage method described above is shown in Figure 1B. Advantages of the method include the following: (i) CCI can be measured directly at the target temperature and not after thawing; (ii) it is semi-destructive, as there is no need to remove the product for measurement but the flip-off cap must be removed in order to puncture the stopper. There

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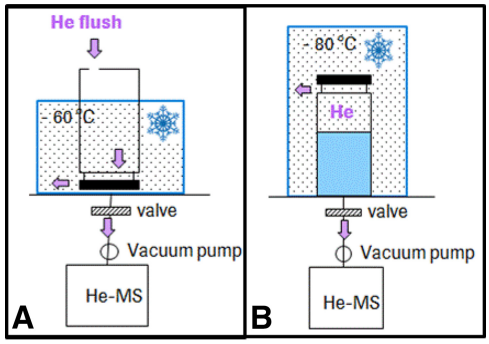

Figure 1

Schematic of He leakage test based methods to measure CCI of frozen drug product: (A) ThermCCI ("outside-in" way) (left); (B) "inside-out" way (right).

are, however, major limitations of this method: (i) the vials are filled with product, which could have an effect on the He dissolved into the liquid; (ii) the He diffusion from the vials (as measurement progresses) may slow down further diffusion due to a decrease in the concentration gradient inside and outside the vial; (iii) the He diffusion should be corrected, as described above, to account for the pressure reduction derived from the low temperatures.

Due to the disadvantages of the existing methods, the development of a new method was pursued without the need to account for pressure correction due to temperature-dependent diffusion. We call this new method *ThermCCI*. The general principle of the ThermCCI method, developed by the authors and described in this article, is to assess the pCCI of a pharmaceutical package by monitoring the He leakage rate directly at the intended storage and/or shipment target temperature (ranging from +40 to –70 °C).

In order to evaluate the ThermCCI method, we assessed pCCI for a variety of CCSs and compared the technology against gas headspace analysis (Figure 1).

Advantages and limitations of the mentioned methods are summarized in Table II.

Materials and Methods

Materials

Type I glass vials (2 mL) (Schott AG, Mainz, Germany), with a 13 mm vial neck diameter were sealed using serum rubber stoppers with a glass transition temperature (T_g) –66.5 °C (Stopper A) and T_g –105 °C (Stopper B). Further details regarding the stoppers are proprietary information and cannot be disclosed. All vials were sealed using a 13 mm crimp cap (Datwyler, Altdorf, Switzerland) made of aluminum and polypropylene.

Artificial leaks were introduced in the samples by inserting uncoated Copper (Cu) wires (Elektrisola Feindraht AG, Escholzmatt, Switzerland) of defined diameter (20 μm and 60 μm) between the rubber stopper and the vial.

Equipment

Vials were sealed with a semi-automatic capping machine (model Milano, Capsulit S.p.A, Roncello, Italy). The sample preparation for the He leakage test involved making two slit-shaped cuts at the bottom of the vial with a glass cutting diamond saw (model 72/800, Arnold, Weilburg, Germany), washing with water, and drying in an oven (model FD-53, Binder GmbH, Tuttlingen, Germany).

An He mass spectrometer (model HLT-560, Pfeiffer Vacuum Technology AG, Zürich, Switzerland) was used for the CCI tests. The He leak detector was prepared for use by conditioning, calibrating, and performing system suitability. NIST (National Institute of Standards and Technology)-traceable He leak cylinders were used. Paraliq GTE 703 (Kluber, Munich, Germany) sealing grease (service temperature –50 to +150 °C) was used in O-ring fittings. A dynamic temperature control system (model Unistat 705w, Peter Huber Kältemaschinenbau GmbH, Offenburg, Germany) was used to control the temperature of the thermo-chamber. A temperature data logger (model Libero Te1P, ELPRO-BUCHS AG, Buchs, Switzerland) monitored the temperature of the vial in the adapter during CCI testing. The temperature probe consisted of a metal bar, and the temperature logging system was recalibrated by ELPRO in compliance with ISO 17025. The thermo-chamber was custom (A-Z Labortechnik, Allschwill, Switzerland).

The X-ray micro computed tomography system (model Skyskan 1172, Bruker, Ettlingen, Germany) was equipped with a Hamamatsu 100/250 X-ray source operating at 59 kV and an 11-megapixel X-ray camera. The system has a 5 μm focal spot size. Projection images were collected as the specimen rotated and were then reconstructed into a cubic array of

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TABLE II
Comparison of Existing Methods for Assessment of CCI at Low Temperatures (Non-shaded = Advantage; Shaded = Inconvenient)

Category	Modified He Leakage Method (ThermCCI)	Modified He Leakage Method	Headspace Analysis (e.g., Frequency Modulated Spectroscopy)
Sensitivity	High sensitivity	High sensitivity	High sensitivity detection possible. It can potentially detect submicron size effects when measuring time allows
			Longer turnaround time (days) or a gas "bombing" procedure might be needed for sensitive detection (15)
Repeatability	Repeatability	Repeatability	For containers with liquid, potential "clogging" with product may prevent oxygen exchange and cause false-negative detections
Reference samples	Ability to use NIST-traceable standards for leakage	Ability to use NIST-traceable standards for leakage	Need to prepare reference samples
Leak size information	Provides fast information on leak size	Provides fast information on leak size	Dependent upon headspace, volume, pressure, or pre-vacuum
Speed	Fast: He is sprayed around the outside of the package ("outside-in" method) (31)	Total measurement time per sample unknown, i.e., time to reach equilibrium in a difussion controlled set up ("inside-out" method)	Fast
Direct/Indirect	Direct measurement	Direct measurement	Indirect measurement (measures changes in internal headspace: O_2, CO_2, water vapor, pressure). No direct assessment of a leak
Technology	He leakage is a well established method	He leakage is a well established method	Based upon frequency-modulated spectroscopy (FMS)
Temperature	Can adjust temperature (and test ramps)	Can adjust temperature (feasibility of ramps not known)	The test is limited to package that is transparent and contains a targeted gas with its concentration changing by leak. The method would not be able to detect leak in a container packaged and stored under atmospheric pressure because a leak in this container under atmospheric pressure would not result in a measurable change in the concentration of gas headspace
Destructive/Non destructive	Destructive (no inline testing, high toxic products)	Non-destructive (although it implies a puncture through the stopper and sealing, hence the flip-off cap needs to be removed and a needle pierced through the seal)	Non-destructive
Sample preparation	Extensive sample preparation	No need to remove product. However, for liquid-filled vials, the possibility exists of He absorption into liquid	Minimal sample preparation required, although it usually requires modified atmosphere packaging
Investment	Higher cost of leakage testing equipment	Higher cost of leakage testing equipment	Lower cost of leakage testing equipment

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TABLE II (continued)

Category	Modified He Leakage Method (ThermCCI)	Modified He Leakage Method	Headspace Analysis (e.g., Frequency Modulated Spectroscopy)
Correlation to microbial ingress	Correlation to mCCI already established, acceptance criteria available	Not available	Lack of correlation to mCCI (because gas permeation is not a direct measure of mCCI)
Other	Combination products/assembled devices that cannot be disassembled without jeopardizing CCI	Not applicable	Not applicable
	Hard surfaces of rubber components, e.g., laminated tip caps, fully laminated component (hypothesis: large number of very small defects which results in high flow rates even if microbiological tightness is proven)	If He difuses out of the vials, this will slow down diffusion	With availability of headspace, with small clareances in headspace, it could detect the liquid
	Background He in the vacuum system	Background He in the vacuum system	Not applicable

volumetric pixels (voxels) with the same edge dimension as the projections.

The oxygen headspace analyses were performed in a headspace analyzer (model FMS-760, Lighthouse Instruments, Charlottesville, VA, USA). The carbon dioxide headspace analyses were performed in a headspace analyzer (model FMS-CO2, Lighthouse Instruments). The headspace analyzers were calibrated using NIST-traceable standards.

Methods

Empty glass vials were filled with He (g) (99.999%, Carbagas AG, Basel, Switzerland) in an upright position using a flush system made in-house. The headspace oxygen levels were checked immediately after filling using an oxygen headspace analyzer. Accepted residual headspace oxygen levels after He filling were below 5% atm O_2.

During the introduction of the artificial leaks, copper (Cu) wires of defined diameter were sandwiched between the rubber stopper and the glass vial (19). Rubber gloves were used to prevent exogenous matter from adhering to vial samples. Vials with artificial leaks were filled with He and crimped after Cu wire insertion.

The vials and stoppers were visually examined for existing defects.

Crimped vials were labeled, placed in upright position in plastic boxes, and stored in dry ice in a polystyrene foam chest under static conditions. Open vials (positive controls, unstoppered) were also stored in dry ice. The temperature of the vials in the foam chest filled with dry ice was –75 ± 5 °C. Crimped vials were also stored at RT as a control.

After 1 week of exposure to the dry ice in the polystyrene foam chest, the vials were allowed to equilibrate to RT for approximately 1 h prior to headspace analysis. The open vials (positive controls) were stoppered immediately after exposure to dry ice.

For each CCS, 10 vials were measured. In some instances, fewer than 10 vials could be measured due to sample unsuitability (i.e., breakage). The vials were tested initially by headspace analysis (after thawing at RT), then by He leakage test at RT (+25 ± 5°C), –20 ± 5 °C, and –60 ± 5 °C. For the He leakage test, samples were thawed and re-frozen at the testing tem-

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perature for a minimum of 12 h before analysis. The exact same vials were analyzed by both techniques.

The general principle of the ThermCCI method is to assess the CCI of a pharmaceutical package by monitoring the He leak rate directly at the target temperature (ranging from +40 to –70 °C) of storage and/or shipment. The method is applicable to all CCS comprised of elastomer stoppers secured to glass containers with crimped aluminum seals. The analysis was performed by an He leak detector equipped with a mass spectrometer. A custom-made thermo-chamber (temperature adjustment module containing a tempering medium) surrounded the container adapter seat. This ensured that the target temperature at the closure end of the container was reached and maintained during leak rate measurement. The thermo-chamber was pre-conditioned at a defined temperature that was adjusted and monitored during measurement by two independent thermometers, one controlling the thermo-chamber temperature and the other in the area in contact with the CCS. For this purpose, the vial was placed upside down in an appropriate adapter. For different vial neck sizes, custom-made vial adapters were available. Prior to measurement, the samples were pre-conditioned at the target temperature in a freezer. Samples were then placed in the vial adapter and analyzed following standard He leakage test method while the temperature was monitored.

This study aimed at evaluating the impact of storage time, T_g of rubber stopper formulation, and artificial leaks on CCI of frozen vials. A description of the parameters chosen in this study is shown in Table III.

Results and Discussion

In a recent study by Zuleger et al. (4), overpressure was observed in vials stored in frozen conditions. More specifically, when vials filled at RT with air (1atm) were taken to a low temperature, their initial headspace shrank and created under-pressure. At temperatures below the T_g, stoppers can lose their elastic properties and integrity can be lost; then cold dense gas from the storage environment can fill the headspace. When warming the container to RT, the stopper regains elasticity and reseals. The cold dense gas is now trapped inside and expands as temperature increases, creating overpressure. In our study, we aimed to recreate a comparable effect with our experimental set-up (Figure 2).

In the first principle feasibility test, the impact of storage time, rubber stopper composition, and artificial leaks on the CCI of frozen vials was studied. The He leakage test was performed at RT, –20 °C, and –60 °C. We compared the results to headspace CO_2 measurements. Empty glass vials filled with He were crimped and stored in dry ice for 1 week. Any change in the headspace conditions would suggest that closure integrity was lost during deep cold storage resulting in headspace gas exchange (Figure 2).

TABLE III
Process Parameters: Number of Samples, Type of Cu Wires, Time and Storage Conditions Tested During the First Principle Feasibility Study, Including Two Different Vial/Stopper Combinations Evaluated

		Storage Temperature	
		Room Temperature	–80°C (dry ice)
Stopper A		1 week	1 week
	Control (no wire)	10	10
	20 μm	10	10
	60 μm	10	10
Stopper B		1 week	1 week
	Control (no wire)	10	10
	20 μm	10	10
	60 μm	10	10

Artificial leaks were introduced with Cu wires of defined diameter (20 μm and 60 μm). The introduction of a 60 μm diameter wire created an artificial leak that led to leaking vials with 100% failure observed. Hence, it was a positive control. In prior experiments, Cu wires in the rage of 20 μm diameter had been identified as the cut-off artificial leak size where microbial ingress was observed during pCCI–mCCI cross correlation (19). Samples without artificial leaks (no Cu wire inserted between vial and stopper) were also used as a control. Additionally, open vials that

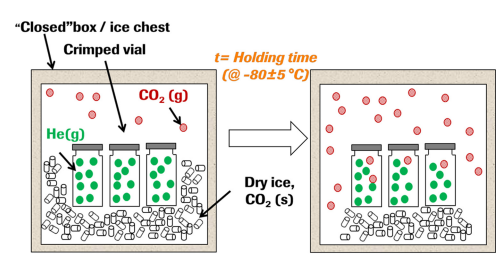

Figure 2

Experimental set up for ThermCCI He leakage method feasibility study.

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TABLE IV
Experimental Runs (Stopper A). Shaded Cells show 1–100% Leaking Vials (Leak)

Storage Conditions	Copper Wire Diameter (μm)	Storage Time (week)	CO_2 Headspace (number of leaking vials/total vials)	He Leak@ –60 °C (number of leaking vials/total vials)	He Leak@ –20 °C (number of leaking vials/total vials)	He Leak@ RT (number of leaking vials/total vials)
RT	No wire	1	0/10	9/10	0/10	0/10
	20	1	0/10	10/10	0/10	0/9
	60	1	3/10*	10/10	10/10	10/10
Dry ice	No wire	1	0/10	3/10	0/10	0/10
	20	1	2/10	9/10	2/10	1/9
	60	1	10/10	10/10	10/10	10/10

* Vials stored in same room as dry ice samples; values measured are in the range of 0.07–0.10% (32).

were open during frozen storage and stoppered were used as control for headspace analysis.

The acceptance criteria were defined as follows: (a) in headspace analysis, all vials showing concentration of CO_2 >0.04% atm (normal concentration of CO_2 in air) were not considered to be tight; (b) in He leakage testing, vials were not considered tight if He leak rate was $\geq 1 \times 10^{-7}$ mbar × L/s.

Table IV shows the experimental results of the pCCI method comparison for Stopper A. The numbers in the table show the number of leaking vials out of the total number of vials tested.

It should be noted that the only comparable results to the headspace CO_2 measurements were the He leakage test results measured at –60 °C in vials stored in dry ice. This is the only tested scenario that could be comparable to the effect of dry ice storage at –80 °C, and that can be observed by CO_2 headspace analysis.

Vials with artificial leaks created by a 60 μm Cu wire diameter, showed 100% leakage by the He leakage test at all test temperatures (RT, –20 °C, and –60 °C) when stored both at RT and in dry ice. Vials with artificial leaks created by a 60 μm Cu wire diameter also showed CO_2 ingress detected by headspace analysis in all cases.

It is worth mentioning that the headspace of vials with artificial leaks introduced by a 60 μm diameter Cu wire showed an abnormally high concentration of CO_2 when stored at RT. In this scenario, three out of 10 vials were identified as having leaked, according to our defined acceptance criteria. It was expected that for these samples the CO_2 content would be within atmospheric levels because they were not exposed to a saturated CO_2 atmosphere in dry ice; however, this was not the case. This observation can possibly be explained by CO_2 ingress from the surroundings. The CO_2 concentration was higher than expected because the vials were stored in the same room where the polystyrene foam chest (containing dry ice) was placed. The CO_2 concentration increase in the storage room atmosphere was also detected by the headspace analyzer as a slightly increased CO_2 permeation through the leak introduced by the Cu wire at RT.

Vials without artificial leaks showed tightness under static conditions both at RT as well as at –20 °C.

The He leakage method at –60 °C showed most of the analyzed vials were leaking after having been previously stored at RT and also when previously stored in dry ice. The He leakage test always detected a higher number of leaking vials as compared to CO_2 headspace analysis. As the same vials were tested by both methods, this observation shows evidence that the He leakage method is more sensitive in detecting leaking vials than CO_2 headspace analysis.

Whenever a vial is thawed after frozen storage, the elastomeric properties of the stopper should be such that every leak potentially having occurred at low temperature would be re-sealed. However, it was observed that in the case of samples with an inserted 20 μm diameter wire (that had been stored in dry ice),1 out of 9 (He leakage at RT) or 2 out of 10 (He leakage at –20 °C) vials showed ingress by the He leakage test, showing they were no longer tight. This was not the case when vials were directly stored at RT, that is, vials were tight at RT. We hypothesize that any potential effect present in the stopper could be enhanced by the low storage temperature, and a memory-like effect could be the underlying cause. More specifi-

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TABLE V
Experimental Runs (Stopper B). Shaded Cells Show 1–100% Leaking Vials (Leak)

Storage Conditions	Copper wire diameter (μm)	Storage Time (week)	CO_2 Headspace (number of leaking vials/total vials)	He Leak at –60 °C (number of leaking vials/total vials)	He Leak at –20 °C (number of leaking vials/total vials)	He Leak at RT (number of leaking vials/total vials)
RT	No wire	1	0/10	9/9	6/9	0/7
	20	1	0/10	10/10	10/10	0/9
	60	1	3/10*	10/10	10/10	10/10
Dry ice	No wire	1	1/10	9/9	1/9	0/10
	20	1	10/10**	10/10	2/10	3/9
	60	1	10/10***	10/10	10/10	9/9

* Vials stored in same room as dry ice samples; values measured are in the range of 0.07–0.10% (32).
** All vials with 20 μm wires exhibit overpressure (>1 atm).
*** All vials with 60 μm wires do not exhibit overpressure.

cally, shrinkage of the stopper could lead to a changed position of the Cu wire, introducing a permanent leak that could result in higher flow rates.

In summary, we found that

- The ThermCCI-He method was more sensitive than the CO_2 headspace method.
- CCS without inserted Cu wires were tight when stored at –20 °C under static conditions.
- CCS without inserted Cu wires, stored in dry ice under static conditions, were not tight when measured by the ThermCCI–He method at –60 °C.
- Frozen CCS exhibited (expected) leakage with copper-wire artificial leaks (60 μm diameter wire).
- With drug product stored in dry ice, an increasing number of frozen vials showed leakage also with a 20 μm wire diameter, suggesting these vials were no longer sufficiently tight.

Table V shows the experimental results of the pCCI method comparison for Stopper B. The numbers in the table show the number of leaking vials out of the total number of vials tested.

In the case of Stopper B studies, all vials with a 20 μm diameter wire stored in dry ice showed overpressure by headspace analysis. CO_2 diffused inside the vials, and after thawing the stopper re-sealed, and the CO_2 trapped inside the vial created overpressure after warm-up and increased the pressure above 1 atm. In contrast, the vials with a 60 μm diameter wire, stored in dry ice, did not show overpressure by headspace analysis. We hypothesize that in this case the size of the originated leak was so large that even upon thawing the stopper was not able to reseal again so the overpressure was released (24).

Vials with artificial leaks created by a 20 μm diameter Cu wire and measured by the He leakage test at several temperatures were not always tight at RT nor at –20 °C when Stopper B was used. This variable result with an inserted 20 μm diameter Cu wire could be explained based on appropriateness of CCS fitting, as discussed below.

All vials stored in dry ice showed leakage when tested under frozen conditions (He leakage test at –60 °C), independently of the inserted artificial leak by Cu wires or not, as observed for stopper A.

The He leakage test method detected 100% leaking vials in all cases at –60 °C (ThermCCI), whereas CO_2 headspace analysis was less sensitive in all cases.

In summary, it was concluded that the ThermCCI–He method was more sensitive than the CO_2 headspace method.

To investigate the root cause for the observed behavior of Stopper B, we further analyzed the crimped vials by computed tomography (CT) and compared them to Stopper A (25).

Stopper B is marketed by the stopper supplier to be specifically designed for frozen drug products, available off-the-shelf. If compression forces are ignored, a tight system is still present, even without crimping (as

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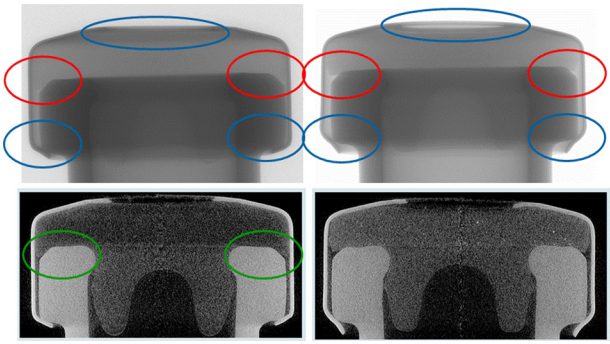

Figure 3

Computer tomography scans of vials crimped with Stopper A (left) and Stopper B (right). Critical sealing areas indicated by circles.

shown by overpressure in control vials that had not been crimped). Hence, this was not an obvious misfit between rubber stopper and glass vial.

Figure 3 shows CT scans of crimped vials, comparing Stopper A and Stopper B side-by-side. Significant differences between both stoppers could be observed. Firstly, a gap on the upper part between the stopper and the flip-off cap could be observed for Stopper B but not for Stopper A, which always showed an acceptable fit. We concluded that the fit of Stopper B and the vial had not been as good as compared to the fit of Stopper A, even if the same crimping machine with the same setting was used. Interestingly, we observed a crease at the edge of the neck of the vial. A gap can be observed in this area (see circles in Figure 3). It was hypothesized that upon freezing, the contraction (shrinkage) of the stopper may lead to a container closure breach at this edge (sealing area). Therefore, it was concluded that differences between vials with Stopper A and those with Stopper B (which cannot be detected by visible inspection) are revealed by CT scans, and that these observed differences could be the potential root cause for differences in CCI behavior.

Other authors also found that a lower stopper T_g does not mitigate CCI failures in worst-case conditions (10, 11). They also concluded that vial/stopper fit and appropriate capping/crimping play crucial roles in CCI and that more studies are needed to fully describe the behavior of lower T_g stoppers.

Two parameters should be considered when analyzing these results:

(a) Dimensions of the Stoppers (Shrinkage). Dimensional analysis of stopper A at different temperature

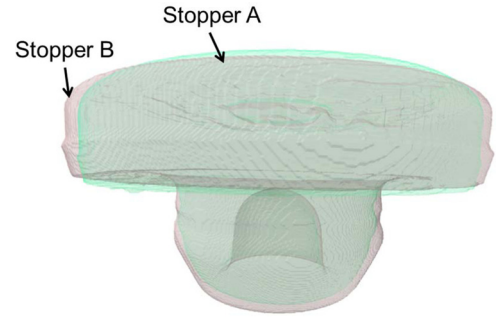

Figure 4

X-ray micro CT scans of overlapped stoppers. Stopper A is shown in green and Stopper B is shown in pink.

(results not shown) indicated that the changes for the glass and the aluminum cap can be negligible. The shrinkage of the stoppers is a few (~10) micrometers at −80 °C. Most probably this is compensated by the pre-crimping pressure due to the stopper being compressed. It is difficult to assess what the shrinkage would look like for the compressed stoppers.

The temperature/shrinkage effect must be seen in combination with the loss of elasticity when the glass transition temperature is reached. Then slight dimensional changes and the loss of elasticity could lead to CCI problems, especially in combination with transportation stress because a hard stopper material could be in contact with a hard glass surface. Inhomogeneities cannot be compensated for. Experimental data for Stopper A suggests that this is not leading to direct problems under static conditions, but transportation stress must be taken into consideration and requires further assessment.

Figure 4 shows an X-ray micro-CT reconstruction of Stopper A and Stopper B. Both scans have been overlapped. Differences can be observed in the stopper dimensions that could explain a misfit of vial/stopper/flip-off cap combination in the case of Stopper B.

(b) Properties of Rubber Stoppers (Rubber-Type Behavior). When the rubber is cooled below its glass transition temperature, it loses elasticity and becomes hard and brittle like glass. If CCI depends on the elasticity of the rubber stopper, one might expect potential sealing issues below the T_g of the rubber stopper. Due to frozen storage being colder than the T_g of most rubber stopper formulations,

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there is an increased risk of losing CCI (26–28). Thus, based on the physical properties of rubber, storage of vials below the T_g has a potential for loss of seal integrity.

Conclusions

The ThermCCI method set-up is able to measure the CCI of frozen vials via an He leakage test method directly at the target temperature.

ThermCCI is a highly sensitive method (direct—in the frozen state) considered more sensitive and suitable than headspace analysis (indirect—upon thawing).

If leaks are present, the impact on CCI appears to be higher at lower storage temperatures. Therefore, selection of the stopper (and fit) is very important.

The CCS tested did not fulfill CCI requirements when stored frozen at –60 °C (under static conditions and assuming the capping process is under control), and in some cases it also failed at –20 °C, depending on the vial/stopper combination. Stopper A, without artificial leaks, was tight at –20 °C. Stopper B was not tight. Thus, appropriateness of frozen drug product storage/shipment should be considered on a case-by-case basis, and CCS combinations need to be individually assessed.

Further studies include broadening the database with more rubber stopper formulations, upgrading the ThermCCI set-up to reach temperatures below –70 °C, performing calculations to assess artificial leak sizes induced by Cu wires with varying diameters, performing calculations to better understand the time-dependent effects of rubber sealability after thawing as well as to understand the potential risk of deformation of the crimp cap and rubber stopper during thawing, monitoring of processing parameters such as residual sealing forces after capping and throughout the process (freezing and thawing), and investigate the effect of capping and shipping stress as well as freeze/thaw cycles on the CCI of frozen drug products.

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Conflict of Interest Declaration

The authors declare that they have no competing interests.

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